

The Weekly Pulse — Multi-Agent Architecture

Agentic AI system for automated newsletter generation · 9 agents · 4 execution phases

Version 2.0 · May 2026

FROZEN

1 DISCOVERY PHASE — ORCHESTRATOR SPAWNS SECTION AGENTS

Orchestrator Agent

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orchestrator

Master coordinator — spawns all agents, manages phase sequencing, handles errors and retries, assembles final newsletter output

`spawn_agent(type, config)` `collect_results(agentIds[])` `route_to_hitl(candidates)` `invoke_renderer(items)` `handle_error(id, err)` `read_memory(key)` `write_memory(key, val)`

↓ spawns in parallel — one instance per newsletter section

SECTION RESEARCH AGENTS — 5 INSTANCES, RUNNING IN PARALLEL

S1 — AI & ML

worker

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Foundational AI/ML research scope

S2 — Dev & OSS

worker

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Developer ecosystem scope

S3 — Industry

worker

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Industry & Gen AI business scope

S4 — Hackathons

worker

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Competitions & events scope

S5 — Data Science

worker

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Data science & analytics scope

Each Section Agent delegates to these two sub-agents sequentially:

Research Subagent

subagent

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Runs 3 search iterations autonomously — decides queries, judges when pool is sufficient, returns 8-15 raw candidates

`web_search(query)` `rss_fetch(feedUrl)` `scrape Og_metadata(url)`

Quality & Curation Agent

subagent

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Checks content integrity, scores on 5 factors, flags [HIGH IMPACT] stories, returns ranked Top 6 JSON

`check_dedup_log(url)` `validate_url(url)` `assess_integrity(article)`

2 HUMAN-IN-THE-LOOP — USER SELECTS FINAL 4 PER SECTION

User Proxy Agent

No LLM — pure interface layer

Presents Top 6 × 5 sections to user via CLI (score + [HIGH IMPACT] flag per candidate) — pauses execution — collects 4 selections per section — returns confirmed 20 items to Orchestrator

human-in-the-loop

```
display_candidates(section, top6[])  collect_stdin_input()  validate_selection(input, count=4)  confirm_final_20(selections)
```

3 GENERATION PHASE — CONTENT AND MEDIA RUN IN PARALLEL

CONTENT & MEDIA AGENTS — PARALLEL ACROSS ALL 20 CONFIRMED ITEMS

Content Synthesis Agent

generator

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Fetches article source, writes a strictly source-faithful 50-word summary — refuses to infer or hallucinate any detail not in the article

```
fetch_article_content(url)  generate_summary(article)  verify_word_count(summary)
verify_source_fidelity(s, a)
```

Media Agent

generator

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Sources representative image per item — OG scrape → Unsplash API → AI-generated fallback — validates 16:9 800×450px

```
scrape_og_image(url)  search_unsplash(kw)  generate_ai_image(prompt)
validate_image_dims(p)  save_image(p, f)
```

4 OUTPUT PHASE — QUALITY GATE THEN RENDER

Rendering Agent

renderer

No LLM — Handlebars template engine

Runs 11 pre-publish quality checks → renders newsletter.html (inline CSS, 600px, Gmail/Outlook-safe) → exports PDF → writes /output/YYYY-MM-DD/ → signals Orchestrator to update Memory Agent

```
run_quality_checks(items[])  render_html_template(items[])  inline_css(html)  export_pdf(html)  write_output_dir(date)  report_status()
```


